

MD. Mehedi Hasan Maruf

Software Engineer

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Summary

Computer Science engineer (RUET) researching NLP-driven curriculum optimization, combining semantic analysis, clustering (K-Means, TF-IDF), and knowledge-graph modeling to surface structural bottlenecks in academic data. Complements this research with almost 4 years of industry experience building production AI systems, including a LangChain-based generative AI agent platform and RAG pipelines, alongside full-stack TypeScript/Python backend development at Aamira Software Solutions. Brings strong algorithmic fundamentals and hands-on DevOps experience (Docker, CI/CD) to research and applied engineering alike.

Skills

Programming Languages: TypeScript, Python, JavaScript

Web Technologies: Angular, React.js, Next.js, Node.js, FastAPI, Django

Databases: MySQL, PostgreSQL, MongoDB, Supabase

Tools & Concepts: Git, Docker, CI/CD (GitHub Actions, Netlify), VPS (Linux) Management, REST APIs, ORM (Django ORM, Supabase), Project Management, Problem Solving

AI/ML: LangChain, Semantic Analysis, NLP, RAG, RLHF Concepts

Experience

Full Stack Developer

Jan 2025 – Present

Aamira Software Solutions 🌐, Kokomo, Indiana, USA

- Spearheaded the frontend development of a high-traffic CRM system using **Angular** and **TypeScript**, directly contributing to increased client engagement.
- Architected and implemented a Typescript microservice backend solution that reduced API response time.
- Optimized database queries using Supabase, resulting in 60% improvement in application performance.

Industrial Trainee

Feb 2025 – Mar 2025

Brain Station 23 🌐, Dhaka, Bangladesh

- Gained practical experience in enterprise software development methodology and practices.
- Served as Team Representative for a 15-person cohort, coordinating communication between management and trainees.
- Contributed to a student AI system using Next.js and Node.js that was deployed to an online server.

Junior Software Engineer

Nov 2022 – Apr 2024

Code Studio 🌐, Rajshahi, Bangladesh

- Developed backends supporting thousands of daily users using JavaScript, TypeScript, and Python (Django).
- Managed 12 VPS (Linux) with 99.8% uptime and optimized CI/CD pipelines, reducing deployment time by 40%.
- Implemented Docker containerization, simplifying deployment across 5 different client environments.

Personal Projects

Generative AI Agent Platform

Feb 2025 – Mar 2025

Repository: [student_AI](#) | Live: [student-ai-next.vercel.app](#)

- Engineered a **Generative AI Agent** using LangChain and LLMs to autonomously create educational content, mirroring the **agent authoring workflows** found in modern AI studios.
- Implemented intelligent quiz generation, document Q&A, and YouTube content analysis using semantic search with conversation memory.
- Built a comprehensive document processing pipeline supporting PDFs, images, and videos with real-time WebSocket chat, JWT authentication, and progress tracking.
- Deployed production-ready application on Vercel (frontend) and Render (backend) with Supabase database integration.

TRACE Framework - AI-Driven Curriculum Optimization (Thesis)

Dec 2025 – Present

- **Architected** a novel NLP pipeline to deconstruct university course syllabi, utilizing advanced semantic analysis to identify structural bottlenecks and curriculum redundancies.
- **Engineered** a specialized algorithm combining **K-Means clustering** and **TF-IDF** to quantify topic relevance, enabling data-driven decisions for academic syllabus restructuring.
- **Designed** a scalable system architecture capable of parsing unstructured academic data into a structured knowledge graph, laying the groundwork for automated educational standard auditing.

Education

B.Sc. Engineering, Computer Science and Engineering

Mar 2022 – Aug 2026

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.77 (till 7th semester)

Certifications & Achievements

- Digital Accessibility Training (DAT) - a2i (Aspire to Innovate), Govt. of Bangladesh [🔗](#) (Oct 2025)
- Industrial Training Certificate - Brain Station 23 [🔗](#) (Mar 2025)
- PHP (Laravel) - EDGE Project [🔗](#) (Feb 2025)
- 4th Place - RUET CodeSmash (Team Competition) - Competitive Programming Contest [🔗](#) (Mar 2023)
- 4th Place - RUET CodeSpark (Individual) - Competitive Programming Contest [🔗](#) (Oct 2022)